

DNN Theory Concepts: Key Explanations for Theoretical Questions

1. Role of Bias in Neural Networks

- Bias allows shifting the activation function to the left or right.
 - It provides the network with an extra degree of freedom, enabling better fitting of data.
 - Mathematically: $z = w \cdot x + b$. Without b , all activations pass through origin.
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2. Vanishing Gradient Problem

- Occurs when gradients become extremely small during backpropagation in deep networks.
 - Affects sigmoid/tanh activations as their derivatives are < 1 , leading to gradients shrinking layer by layer.
 - Solution: Use ReLU, He Initialization, Batch Normalization, or Residual Connections.
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3. Difference Between Binary and Multiclass Classification

Task Type	Activation Function	Loss Function
Binary	Sigmoid	Binary Cross-Entropy
Multiclass	Softmax	Categorical Cross-Entropy

- Sigmoid outputs independent probabilities; Softmax normalizes across classes.
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4. Dropout Regularization

- Technique to prevent overfitting by randomly setting a fraction of inputs to zero during training.
 - Introduces noise, reduces co-adaptation.
 - During inference, all units are active but scaled.
 - Number of thinned networks: 2^n (n = number of units).
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5. Weight Initialization Strategies

Method	Formula	Use Case
Xavier	$\sqrt{1/n}$	Sigmoid/Tanh
He Init	$\sqrt{2/n}$	ReLU

- Good initialization prevents vanishing/exploding gradients.

6. Adam Optimizer Parameters

- β_1 : Controls momentum (usually 0.9) $\rightarrow$ average of past gradients.
 - β_2 : Controls RMS component (usually 0.999) $\rightarrow$ average of squared gradients.
 - Combines momentum + adaptive learning rate.
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7. Transfer Learning

- Pretrained models are reused for new tasks (usually on smaller data).
 - Advantages:
 - Faster convergence
 - Prevents overfitting
 - Utilizes previously learned features
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8. Generalization vs Overfitting

- Overfitting: High training accuracy, low test accuracy
 - Causes: High capacity model, less data
 - Solutions: Dropout, Early Stopping, Regularization, Data Augmentation
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9. Activation Functions Overview

Name	Formula	Range	Remarks
Sigmoid	$1 / (1 + e^{(-x)})$	(0,1)	Vanishing gradients
Tanh	$(e^x - e^{-x}) / (e^x + e^{-x})$	(-1,1)	Zero-centered
ReLU	$\max(0, x)$	$[0, \infty)$	Fast, sparse gradients
Leaky ReLU	$\max(\alpha x, x)$	$(-\infty, \infty)$	Avoids dying ReLU
Softmax	$e^{z_j} / \sum e^{z_k}$	[0,1]	For classification

10. Batch Normalization

- Applied after linear transformation but before activation.
- Normalizes input per batch: mean = 0, variance = 1
- Benefits:
 - Stabilizes training
 - Allows higher learning rates
 - Slight regularization effect

Let me know if you'd like exam-style Q&A added from each topic or need visual diagrams.